

# List of publication and products

Prof. Dr. Pierros Ntelis

June 8, 2026

Via, a 14 *h*-index, a 16 *i10*-index, and total citations: 4002 ([Google Scholar 2026](#)), my most important work is represented in the following papers:

## Accepted

- [1] P. Yovkochev and **Ntelis, P.**, 2026. Evolution and Mass Decay of a Gaussian Pulse in Time-Fractional Diffusion: An L1-Implicit Scheme Analysis. *Turanian Journal*, vol. 2, no. 1, pp. 020103-1–020103-7. [URL](#), Contribution: Supervision, and co-design of analysis. (Citations: 0)
- [2] **Ntelis, P.** 2025. Advancing Tensor Theories. Symmetry: Mathematics: Advances in Topology and Algebraic Geometry. [URL](#) (Citations: 2) ★
- [3] **Ntelis, P.** 2025. Advanced manifold-metric pairs. Mathematics. [ArXiv](#), [URL](#) (Citations: 1) ★
- [4] **Ntelis, P.** & J. L. Said, 2025. Analytical *poly* $\Lambda$ CDM dynamics. PDU.[URL](#) (Citations: 1) ★
- [5] E.D. Valentino, J. L. Said, ..., **Ntelis, P.** ..., et al. 2025. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. Contribution: Writing on extended theories of gravity, scientific review, and suggestions for further development. PDU [URL](#) (Citations: 472)
- [6] **Ntelis, P.** 2022. A  $(D_\tau, D_x)$ -manifold with  $N$ -correlators of  $N_t$ -objects. IJGMMP, arXiv e-prints arXiv:2209.07472 [URL](#), (Citations: 0)
- [7] **Ntelis, P.** & J. L. Said 2025. Exploring  $\phi$ CDM model dynamics. EPJC, arXiv:2502.03486v1 [URL](#), (Citations: 4)
- [8] **Ntelis, P.** & J. L. Said 2025. Simple  $\phi\Lambda$ CDM dynamics. IJGMMP, arXiv:2502.03486v2 [URL](#), (Citations: 3)
- [9] S.Casas, ..., **Ntelis, P.**, and 128 colleagues 2025. Euclid: Constraints on  $f(R)$  cosmologies from the spectroscopic and photometric primary probes. A&A, [URL](#) (Citations: 27) Contribution: Scientific review, software development, and contribution on discussions for further research.
- [10] Euclid Collaboration, Y. Mellier and 1137 colleagues 2024. Euclid. I. Overview of the Euclid mission. A&A, arXiv:2405.13491 [URL](#), (Citations: 785) Contribution: Scientific review and contribution on discussions for further research.
- [11] **Ntelis, P.** & Morris, A. 2023. Functors of actions. Foundations of Physics, arXiv:2010.06707 [URL](#), (Citations: 24) ★
- [12] Göksel Karaçaylı, N., ... **Ntelis, P.**, and 10 colleagues 2022. Optimal 1D  $\text{Ly}\alpha$  Forest Power Spectrum Estimation – II. KODIAQ, SQUAD & XQ-100. MNRAS [URL](#), (Citations: 58), Contribution: Scientific review and contribution on discussions for further research and software development.
- [13] Hamaus, N., ... **Ntelis, P.**, and 128 colleagues 2022. Euclid: Forecasts from redshift-space distortions and the Alcock-Paczynski test with cosmic voids. A&A arXiv e-prints [URL](#), (Citations: 95), Contribution: Scientific review and contribution on discussions for further research and software development.
- [14] Ilić, S., ... **Ntelis, P.**, and 207 colleagues 2021. *Euclid* preparation: XV. Forecasting cosmological constraints for the *Euclid* and CMB joint analysis. A&A arXiv e-prints [URL](#), (Citations: 104), Contribution: Scientific review and contribution on discussions for further research. ★
- [15] Alam, S., ... **Ntelis, P.**, and 33 colleagues 2020. Towards testing the theory of gravity with DESI: summary statistics, model predictions and future simulation requirements. JCAP [URL](#),

- (Citations: 75), Contribution: Scientific review and contribution on discussions for further research and software development. ★
- [16] Tutusaus, I., ... **Ntelis, P.**, and 89 colleagues 2020. Euclid: The importance of galaxy clustering and weak lensing cross-correlations within the photometric Euclid survey. *Astronomy and Astrophysics* 643, [URL](#), (Citations: 92), Contribution: Scientific review and contribution on discussions for further research.
- [17] **Ntelis, P.**, A. Ealet, S. Escoffier, J.-C. Hamilton, A. J. Hawken, J.-M. Le Goff, J. Rich, and A. Tilquin 2018. The scale of cosmic homogeneity as a standard ruler. *Journal of Cosmology and Astroparticle Physics* 2018, [URL](#), (Citations: 28)
- [18] Blanton, M. R., ... **Ntelis, P.**, and 361 colleagues 2017. Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe. *The Astronomical Journal* 154, [URL](#), (Citations: 2122), Contribution: Scientific review and contribution on discussions for further research. ★
- [19] **Ntelis, P.**, and 22 colleagues 2017. Exploring cosmic homogeneity with the BOSS DR12 galaxy sample. *Journal of Cosmology and Astroparticle Physics* 2017, [URL](#), (Citations: 113) ★
- [20] Laurent, P., ... **Ntelis, P.**, and 17 colleagues 2016. A  $14 h^{-3} \text{ Gpc}^3$  study of cosmic homogeneity using BOSS DR12 quasar sample. *Journal of Cosmology and Astroparticle Physics* 2016, [URL](#), (Citations: 99), Contribution: Scientific review and contribution on discussions for further research and software development.

## Under review

- [1] **Ntelis, P.**, et al. 2019. Cosmological constraints from cosmic homogeneity. (Cites: 4) [URL](#)
- [2] **Ntelis, P.** 2025. A probabilistic and expanding Universe. [URL](#), (Cites: 2)
- [3] **Ntelis, P.** 2023. Advancing category and tensor theories. [URL](#), (Cites: 2)
- [4] **Ntelis, P.** 2025. CosmoFATs. [URL](#), (Cites: 2)
- [5] **Ntelis, P.** 2025. CosmoFATs Dynamics. [URL](#) (Cites: 0)
- [6] **Ntelis, P.** 2025. A cosmic Schiff's conjecture test. [URL](#), , [DOI](#), (Cites: 0)
- [7] **Ntelis, P.** 2025. A universal foundational theory. [URL](#), (Cites: 0)
- [8] **Ntelis, P.** 2026. · A dot. [URL](#), [DOI](#), (Cites: 0)
- [9] **Ntelis, P.** 2026. Analytical sub-poly $\Lambda$ CDM through cosmic obs. [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.** 2026. A foundational analysis of local functional calculus [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.**, B.J. Ahmedov, C.Yuan, 2026. A cosmic femtolensing Hawking radiation signature [URL](#), [DOI](#), (Cites: 0)
- [12] **Ntelis, P.**, 2026. [EinsteinKPy2Tex](#): A Python Tool for Computing and Exporting General Relativity Tensors to LaTeX. signature [URL](#), (Cites: 0)
- [13] **Ntelis, P.**, 2026. A temporal-spatial-entropic metric and its Kretschmann scalar [URL](#), (Cites: 0)

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| <b>Outreach</b> | 2022- $n_{ow}$ , <a href="#">Universions Youtube Channel</a>                                          |
|                 | 2025, <a href="#">Utube</a> , Comet 3i ATLAS: Interstellar Halloween Secrets! (6')                    |
|                 | 2025, <a href="#">Utube</a> , Are there Aliens ? Drake's Equation ! (7')                              |
|                 | 2025, <a href="#">Utube</a> , Quantum Tunnelling in Your Palm! (1')                                   |
|                 | 2025, <a href="#">Utube</a> , Advanced Manifold-Metric Pairs (10')                                    |
|                 | 2025, <a href="#">Utube</a> , A sunny watery magic (33")                                              |
|                 | 2025, <a href="#">Utube</a> , Advanced Euler's identity (3')                                          |
|                 | 2025, <a href="#">Utube</a> , Analytical <i>poly</i> $\Lambda$ CDM dynamics 101 (5' talk)             |
|                 | 2025, <a href="#">Utube</a> , Advancing Tensor Theories 101 (10' talk)                                |
|                 | 2025, <a href="#">Utube</a> , Quintessence 101 (5' talk)                                              |
|                 | 2024, <a href="#">Utube</a> , Introduction cosmology for everyone (1h talk)                           |
|                 | 2023('24), <a href="#">IBS of Provence</a> , Neuroscience and cosmology for high school students (3h) |
|                 | 2023, <a href="#">IBS of Provence</a> , Math and physics from CPT for high school students (3h)       |
|                 | 2022, <a href="#">IBS of Provence</a> , Mathematics and physics for high school students (3h)         |

|               |                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------|
| Conferences   | 2026, <a href="#">HIT, SoP, Advanced math in cosmology</a> (1h Seminar)                                |
| Talks/Posters | 2026, <a href="#">HIT, SoP, A note on Baye's data analysis</a> (1h Seminar)                            |
| Seminars      | 2025, <a href="#">The Dark Side of the Universe, FAT dynamics to cosmology</a> (25' Talk)              |
|               | 2025, <a href="#">Cosmo From Home, Advanced math in cosmology</a> (Talk)                               |
|               | 2024, <a href="#">50 yr of Horndeski at PI, A probabilistic expanding universe under FA</a> , (Poster) |
|               | 2024, <a href="#">Cosmo From Home, A probabilistic expanding universe under FA</a> , (25 min talk)     |
|               | 2023, <a href="#">NEB, New avenues of functors of actions theories</a> , (Talk 20')                    |
|               | 2023, <a href="#">Cosmo From Home, New avenues on Functors of Actions</a> , (25 min talk)              |
|               | 2023, <a href="#">NTUA HET, New theories in cosmology and particle physics</a> , (1h Seminar)          |
|               | 2022, <a href="#">CERN TH DPT Visit, Functors of Actions</a> (5' Blackboard Talk)                      |
|               | 2021, <a href="#">Euclid Science Talk, Overview of impact of interlopers on PNG</a> (1h Talk)          |
|               | 2020, <a href="#">Atelier D.E., PNG with interlopers in Euclid</a> , (20' Talk) HVT                    |
|               | 2020, <a href="#">Atelier D.E., Perturbation Theory of LSS, code: CLASS-PT</a> , (20' Talk)            |
|               | 2020, <a href="#">SEMFE, NTUA, LSS: Theory and Observations</a> , (40' Seminar)                        |
|               | 2019, <a href="#">30th SYMPOSIUM, Fractality from Multiple Tracers</a> , (15' Talk)                    |
|               | 2019, <a href="#">Euclid IAS Paris, Systematics on PNG</a> , (20' Talk)                                |
|               | 2019, <a href="#">Euclid Consortium Helsinki, Primordial Non-Gaussianity</a> , (15' Talk)              |
|               | 2019, <a href="#">eBOSS/DESI France, Cosmology with Cosmic Homogeneity</a> (15' Talk)                  |
|               | 2019, <a href="#">Aix-University, Homogeneity with Multitracers</a> , (2h Seminar)                     |
|               | 2019, <a href="#">Aix-University, A note on lss, Summer School</a> , (6h Seminar)                      |
|               | 2018, <a href="#">Euclid GC-WL, Ameliorating cosmology with homogeneity</a> , (15' Talk)               |
|               | 2018, <a href="#">Colloquium of D.E., Cosmic Homogeneity as a standard ruler</a> , (20' Talk)          |
|               | 2018, <a href="#">R. Moriond, Euclid: Homogeneity in search of the Dark Sector</a> , (Poster) HIC      |
|               | 2018, <a href="#">UiO, What can fractality tell us about the universe?</a> , (1h seminar)              |
|               | 2017, <a href="#">P.O.N.T. in Cosmology, Is the universe homogeneous?</a> , (20' Talk)                 |
|               | 2016, <a href="#">R. Moriond, The homogeneity scale of the universe</a> , (15' Talk)                   |
|               | 2015, <a href="#">UnivEarthS School, Santorini, LSS and homogeneity</a> , (Poster)                     |
|               | 2015, <a href="#">Rencontres de Jeunes Physicienne, Homogeneity (Video)</a> , (15' Talk)               |
|               | 2014, <a href="#">Elbereth Conference, IAP, The homogeneity of the universe</a> (15' Talk)             |
|               | 2014, <a href="#">UnivEarthS School, The homogeneity of the universe</a> , (3' Talk)                   |

## Software

- [1] P.Ntelis (2017-today): [Github contributions](#)
- [2] P.Ntelis (2017-today): [COSMOPIT](#)
- [3] P.Ntelis (2018-2021): [Gitlab private Euclid softwares](#)
- [4] P.Ntelis (2018): [MCMC-posterior-example](#)
- [5] P.Ntelis (2018): [CoHo](#)
- [6] P.Ntelis (2018): [ANALYSE-SN-magnitude-redshift-curve](#)
- [7] P.Ntelis (2018): [Extract-Cosmology-with-the-BA0-peak-position](#)
- [8] P.Ntelis (2020): [AofEFT](#)
- [9] P.Ntelis (2023): [Tutorial on galaxy simulations and coordinate transformation](#)
- [10] P.Ntelis (2025a): [Standard  \$\phi\$ CDM dynamics](#)
- [11] P.Ntelis (2025b): [Simple quintessence dynamics](#)
- [12] P.Ntelis (2025c): [Analytical  \$poly\Lambda\$ CDM dynamics](#)
- [13] P.Ntelis (2026): [EinsteinKPy2Tex](#)